Lab Experiments
Set:11-15
Date:22-07-26

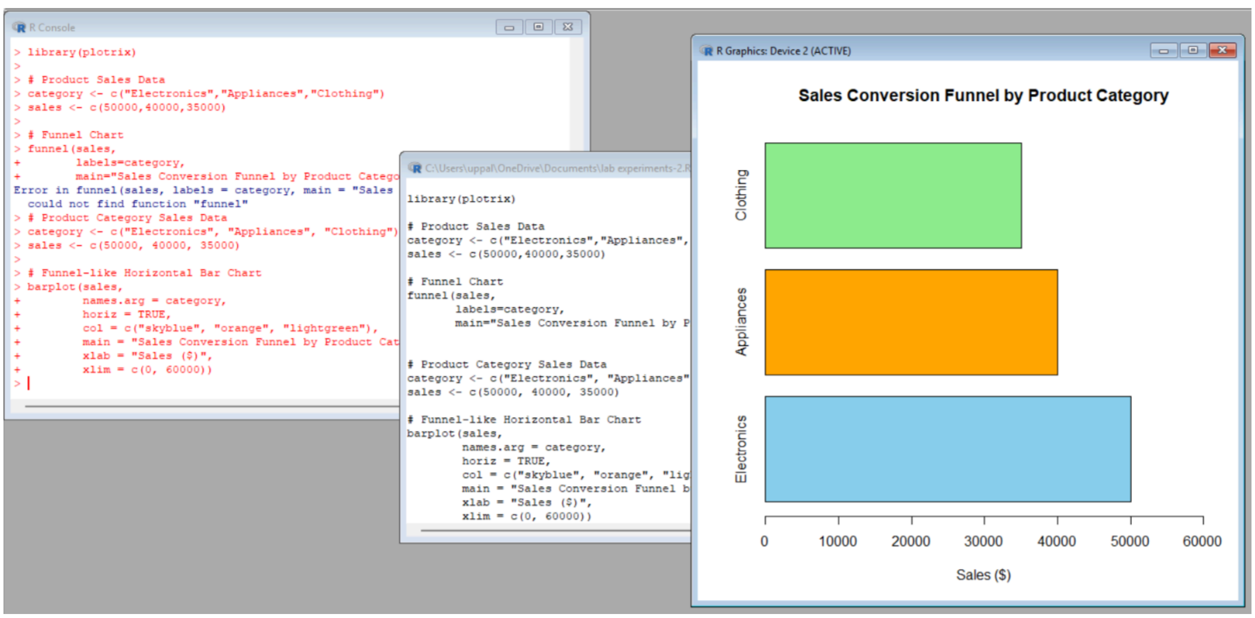
11.Product Category Analysis

1: Funnel Chart

Code:

```
category <- c("Electronics", "Appliances", "Clothing")
sales <- c(50000, 40000, 35000)
barplot(sales,
        names.arg = category,
        horiz = TRUE,
        col = c("skyblue", "orange", "lightgreen"),
        main = "Sales Conversion Funnel by Product Category",
        xlab = "Sales ($)",
        xlim = c(0, 60000))
```

Output:



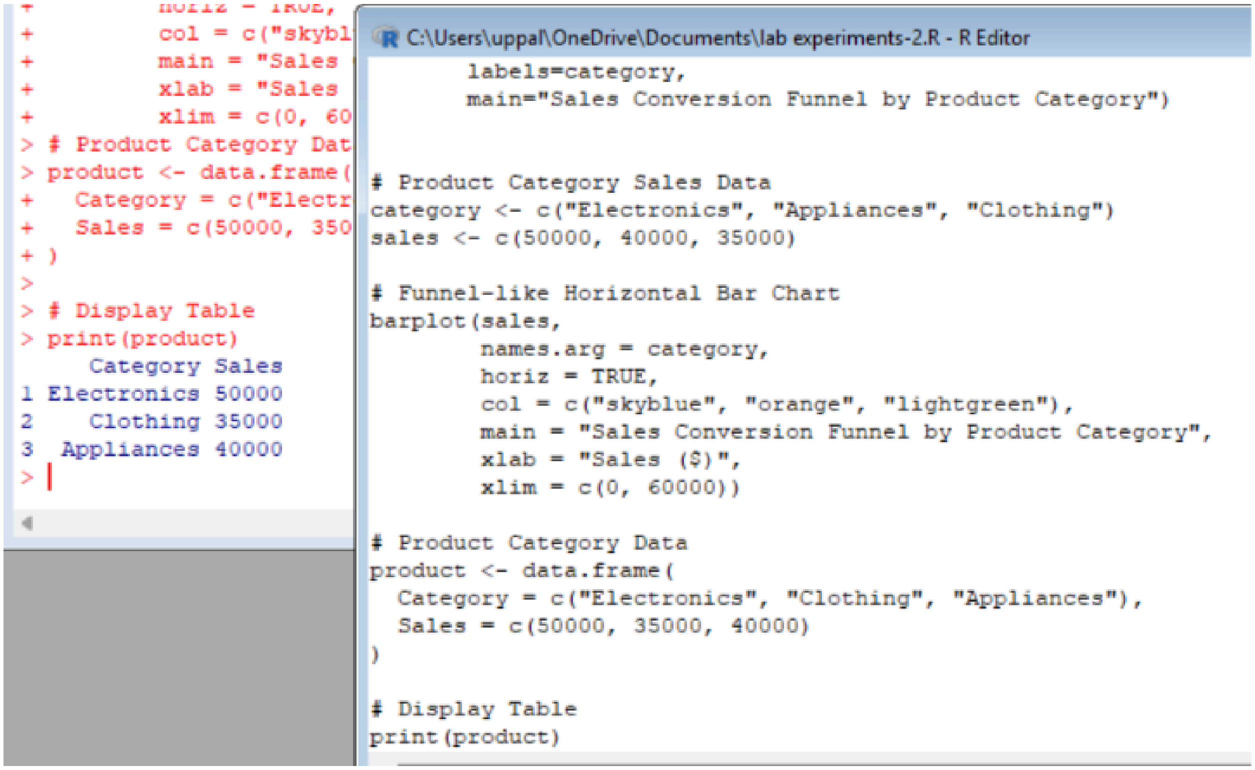
2: Table

Code:

```
product <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
```

print(product)

Output:



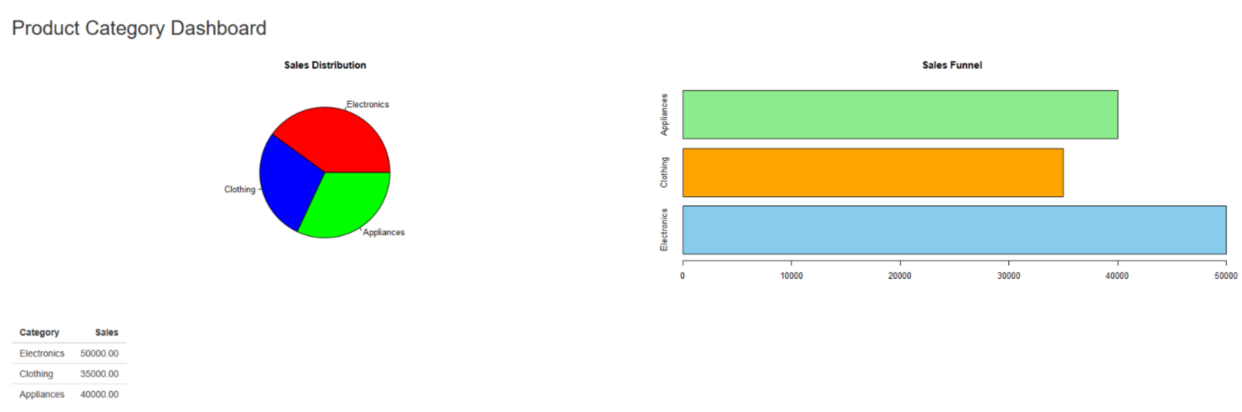
3: Dashboard (R Shiny)

Code:

```
library(shiny)
product <- data.frame(
  Category = c("Electronics","Clothing","Appliances"),
  Sales = c(50000,35000,40000)
)
ui <- fluidPage(
  titlePanel("Product Category Dashboard"),
  fluidRow(
    column(6, plotOutput("pie")),
    column(6, plotOutput("bar"))
  ),
  br(),
  tableOutput("table")
)
server <- function(input, output){
  output$pie <- renderPlot({
```

```
  pie(product$Sales,
      labels = product$Category,
      col = c("red","blue","green"),
      main = "Sales Distribution")
 })
 output$bar <- renderPlot({
  barplot(product$Sales,
          names.arg = product$Category,
          horiz = TRUE,
          col = c("skyblue","orange","lightgreen"),
          main = "Sales Funnel")
 })
 output$table <- renderTable({
  product
 })
}
shinyApp(ui, server)
```

Output:

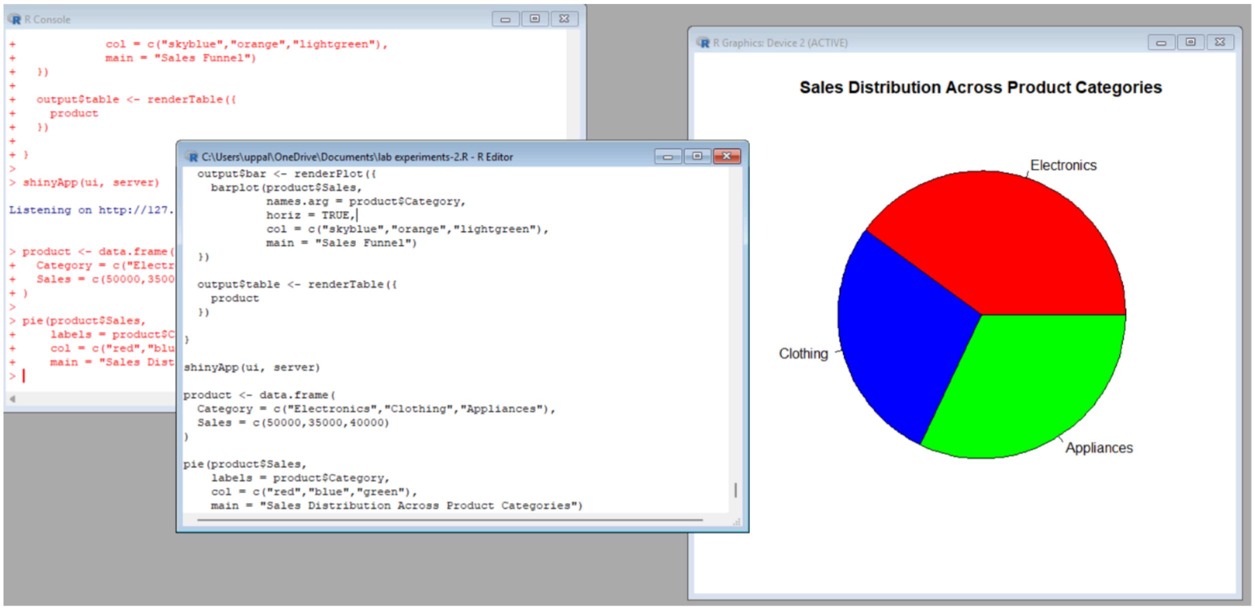


4: Pie Chart

Code:

```
product <- data.frame(
 Category = c("Electronics","Clothing","Appliances"),
 Sales = c(50000,35000,40000)
)
pie(product$Sales,
    labels = product$Category,
    col = c("red","blue","green"),
    main = "Sales Distribution Across Product Categories")
```

Output:



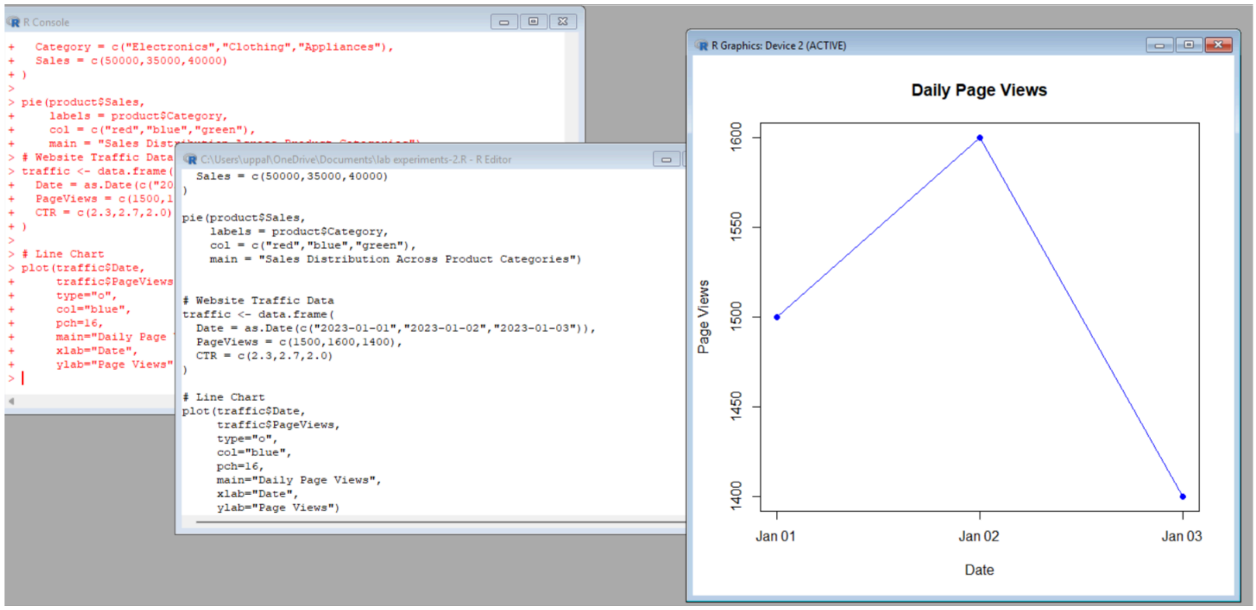
Set 12 :Dataset: Website Traffic

Question 1: Line Chart (Daily Page Views)

Code:

```
traffic <- data.frame(
  Date = as.Date(c("2023-01-01","2023-01-02","2023-01-03")),
  PageViews = c(1500,1600,1400),
  CTR = c(2.3,2.7,2.0)
)
plot(traffic$Date,
     traffic$PageViews,
     type="o",
     col="blue",
     pch=16,
     main="Daily Page Views",
     xlab="Date",
     ylab="Page Views")
```

Output:

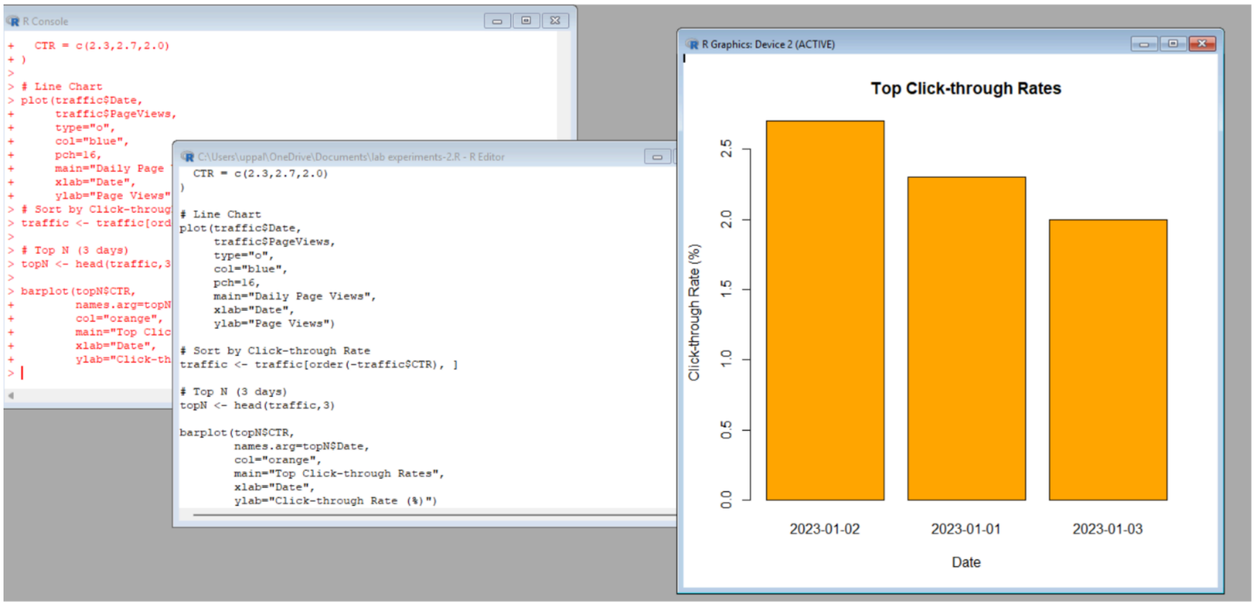


2: Bar Chart (Top N Click-through Rates)

Code:

```
traffic <- traffic[order(-traffic$CTR), ]
topN <- head(traffic,3)
barplot(topN$CTR,
        names.arg=topN$Date,
        col="orange",
        main="Top Click-through Rates",
        xlab="Date",
        ylab="Click-through Rate (%)")
```

Output:

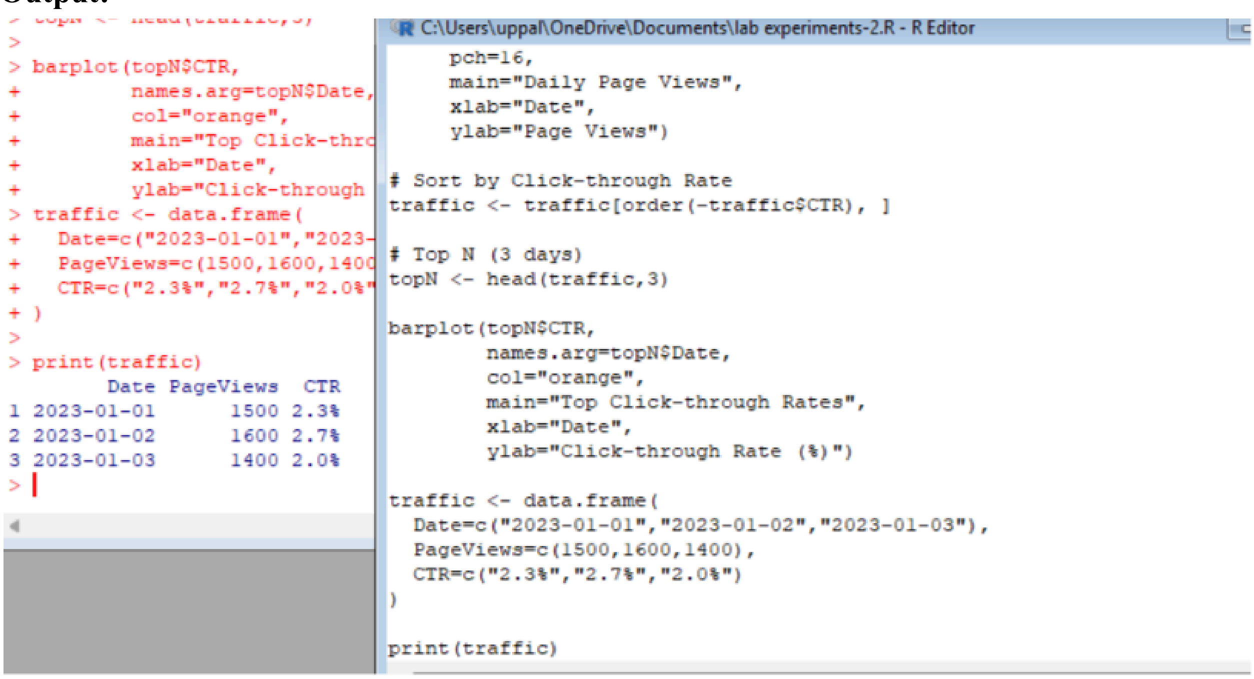


3: Table

Code:

```
traffic <- data.frame(
  Date=c("2023-01-01","2023-01-02","2023-01-03"),
  PageViews=c(1500,1600,1400),
  CTR=c("2.3%","2.7%","2.0%")
)
print(traffic)
```

Output:

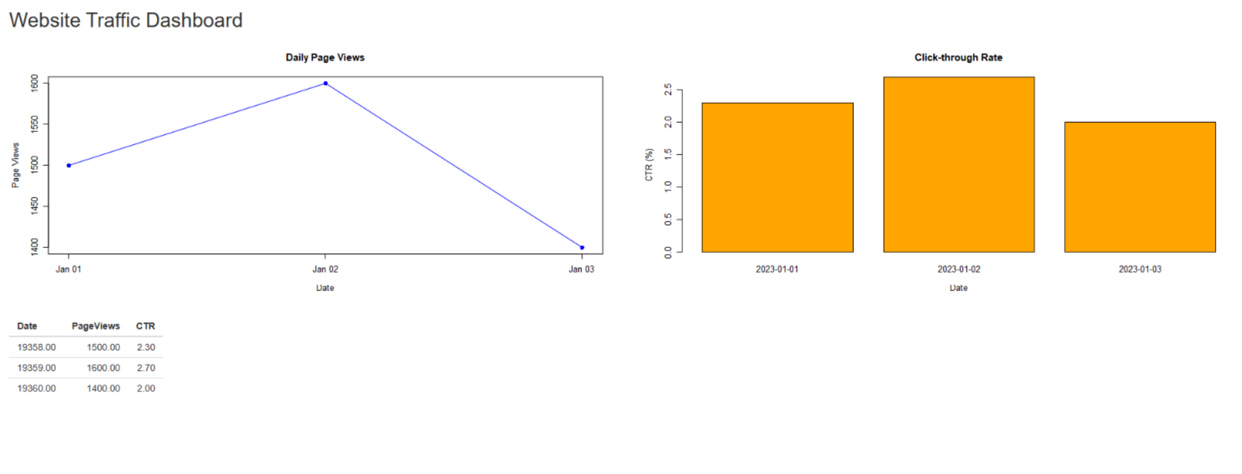


4: Dashboard (R Shiny)
Code:

```
library(shiny)
traffic <- data.frame(
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03")),
  PageViews=c(1500,1600,1400),
  CTR=c(2.3,2.7,2.0)
)
ui <- fluidPage(
  titlePanel("Website Traffic Dashboard"),
  fluidRow(
    column(6, plotOutput("line")),
    column(6, plotOutput("bar"))
  ),
  br(),
  tableOutput("table")
)
server <- function(input, output){
  output$line <- renderPlot({
    plot(traffic$Date,
         traffic$PageViews,
         type="o",
         col="blue",
         pch=16,
         main="Daily Page Views",
         xlab="Date",
         ylab="Page Views")
  })
  output$bar <- renderPlot({
    barplot(traffic$CTR,
            names.arg=traffic$Date,
            col="orange",
            main="Click-through Rate",
            xlab="Date",
            ylab="CTR (%)")
  })
  output$table <- renderTable({
    traffic
  })
}
```

shinyApp(ui, server)

Output:



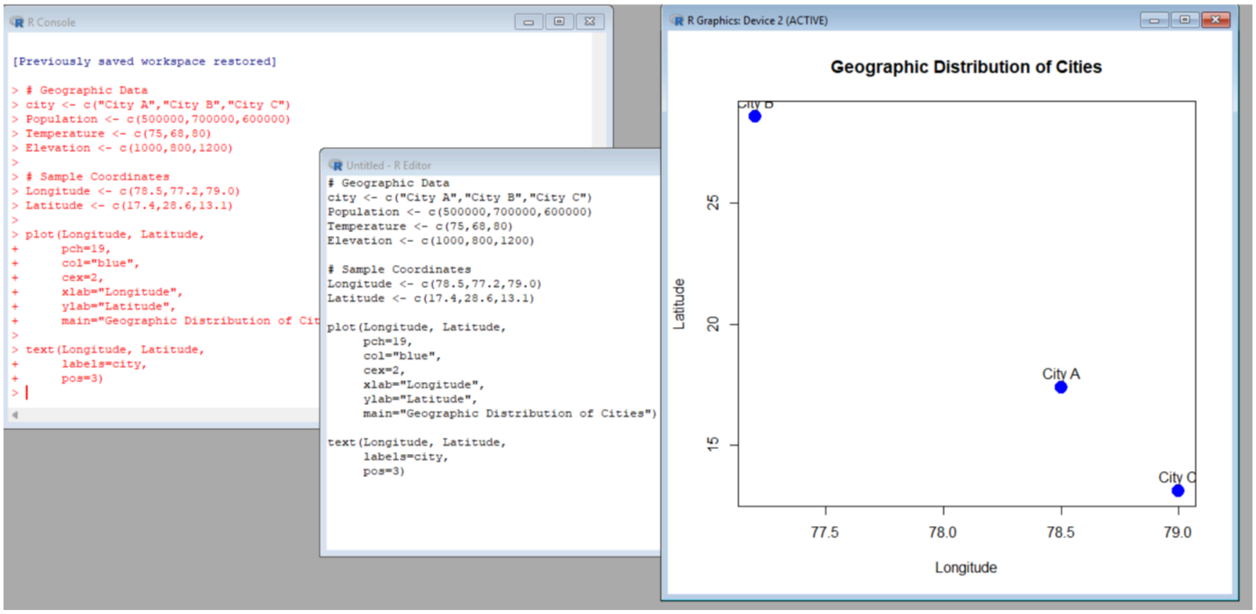
Set13. Dataset: Geographic Data

1: Map Chart (Using Scatter Plot as City Map)

Code:

```
# Geographic Data
city <- c("City A","City B","City C")
Population <- c(500000,700000,600000)
Temperature <- c(75,68,80)
Elevation <- c(1000,800,1200)
# Sample Coordinates
Longitude <- c(78.5,77.2,79.0)
Latitude <- c(17.4,28.6,13.1)
plot(Longitude, Latitude,
    pch=19,
    col="blue",
    cex=2,
    xlab="Longitude",
    ylab="Latitude",
    main="Geographic Distribution of Cities")
text(Longitude, Latitude,
    labels=city,
    pos=3)
```

Output:

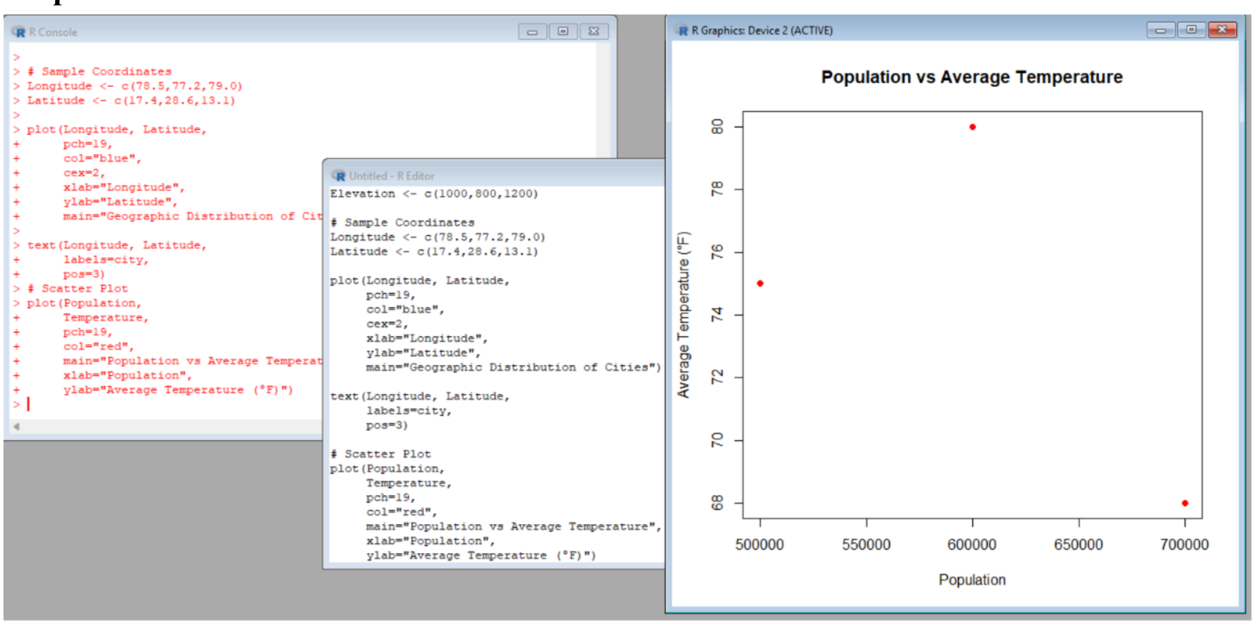


2: Scatter Plot

Code:

```
plot(Population,
     Temperature,
     pch=19,
     col="red",
     main="Population vs Average Temperature",
     xlab="Population",
     ylab="Average Temperature (°F)")
```

Output:

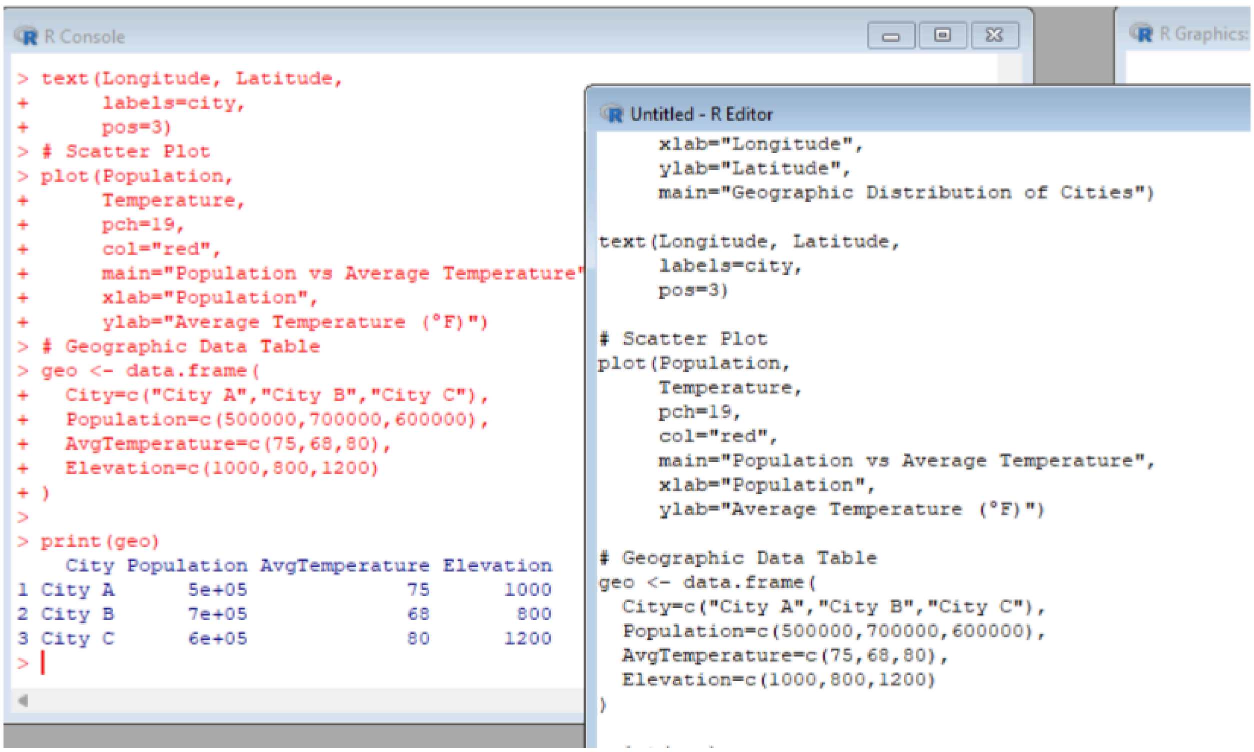


3: Table

Code:

```
geo <- data.frame(
  City=c("City A","City B","City C"),
  Population=c(500000,700000,600000),
  AvgTemperature=c(75,68,80),
  Elevation=c(1000,800,1200)
)
print(geo)
```

Output:



4: Dashboard (R Shiny)

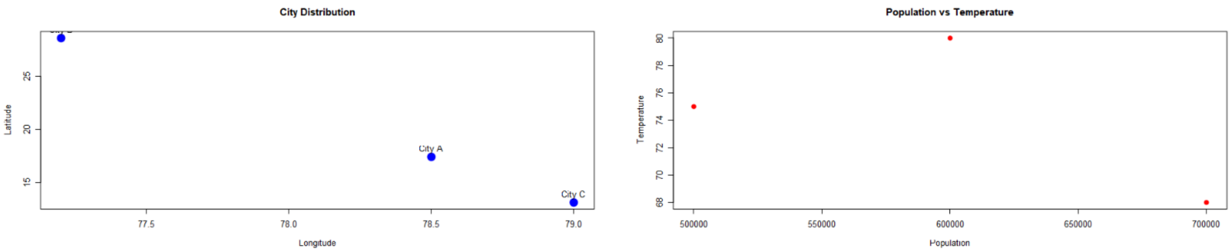
Code:

```
library(shiny)
geo <- data.frame(
  City=c("City A","City B","City C"),
  Population=c(500000,700000,600000),
  AvgTemperature=c(75,68,80),
  Elevation=c(1000,800,1200),
  Longitude=c(78.5,77.2,79.0),
  Latitude=c(17.4,28.6,13.1)
)
ui <- fluidPage(
```

```
  titlePanel("Geographic Data Dashboard"),
  fluidRow(
    column(6, plotOutput("map")),
    column(6, plotOutput("scatter"))
  ),
  br(),
  tableOutput("table")
)
server <- function(input, output){
  output$map <- renderPlot({
    plot(geo$Longitude,
         geo$Latitude,
         pch=19,
         col="blue",
         cex=2,
         xlab="Longitude",
         ylab="Latitude",
         main="City Distribution")
    text(geo$Longitude,
         geo$Latitude,
         labels=geo$City,
         pos=3)
  })
  output$scatter <- renderPlot({
    plot(geo$Population,
         geo$AvgTemperature,
         pch=19,
         col="red",
         xlab="Population",
         ylab="Temperature",
         main="Population vs Temperature")
  })
  output$table <- renderTable({
    geo[,1:4]
  })
}
shinyApp(ui, server)
```

Output:

Geographic Data Dashboard

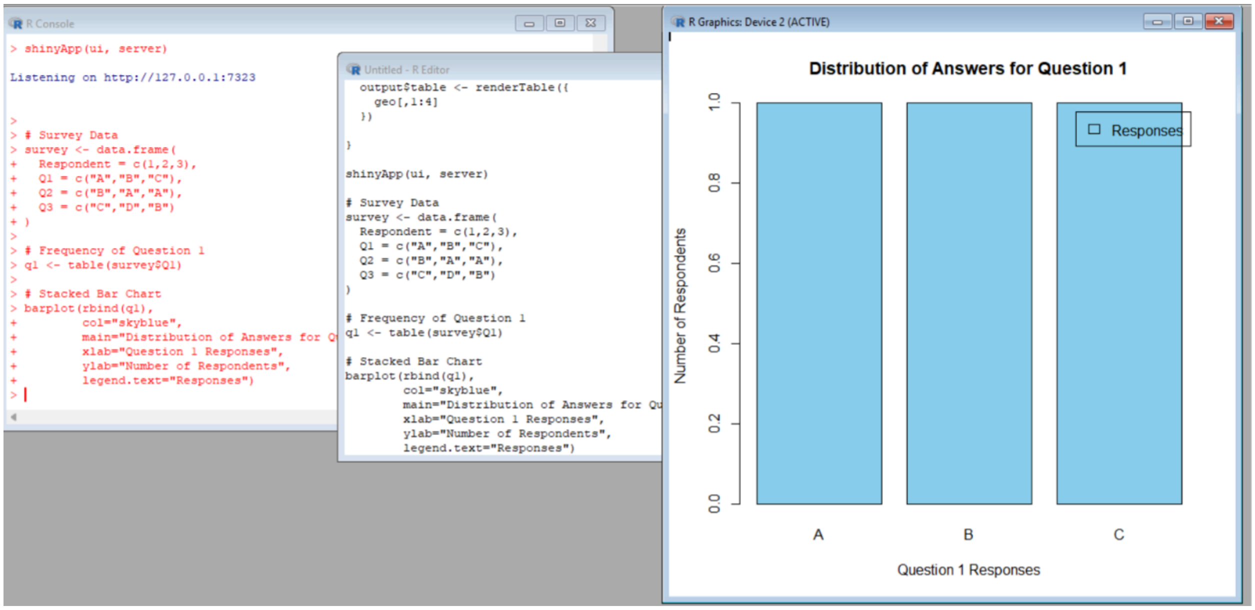


City	Population	AvgTemperature	Elevation
City A	500000.00	75.00	1000.00
City B	700000.00	68.00	800.00
City C	600000.00	80.00	1200.00

14. Dataset: Survey Results
1: Stacked Bar Chart
Code:

```
survey <- data.frame(
  Respondent = c(1,2,3),
  Q1 = c("A","B","C"),
  Q2 = c("B","A","A"),
  Q3 = c("C","D","B")
)
q1 <- table(survey$Q1)
barplot(rbind(q1),
        col="skyblue",
        main="Distribution of Answers for Question 1",
        xlab="Question 1 Responses",
        ylab="Number of Respondents",
        legend.text="Responses")
```

Output:

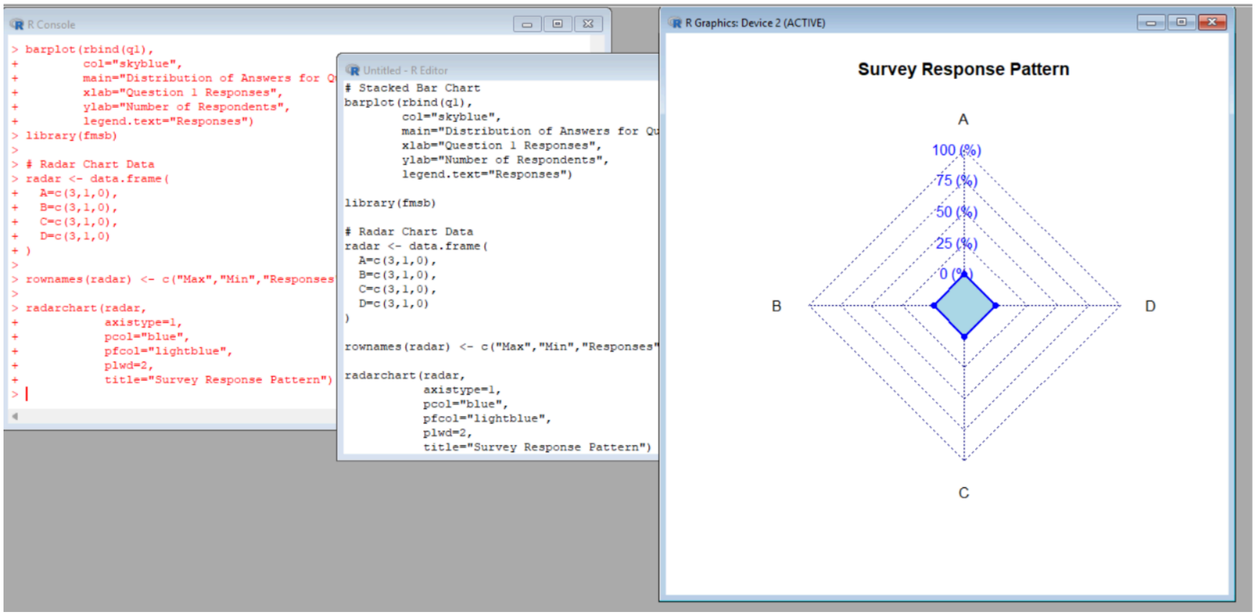


2: Radar Chart

Code:

```
library(fmsb)
radar <- data.frame(
  A=c(3,1,0),
  B=c(3,1,0),
  C=c(3,1,0),
  D=c(3,1,0)
)
rownames(radar) <- c("Max","Min","Responses")
radarchart(radar,
           axistype=1,
           pcol="blue",
           pfcol="lightblue",
           plwd=2,
           title="Survey Response Pattern")
```

Output:

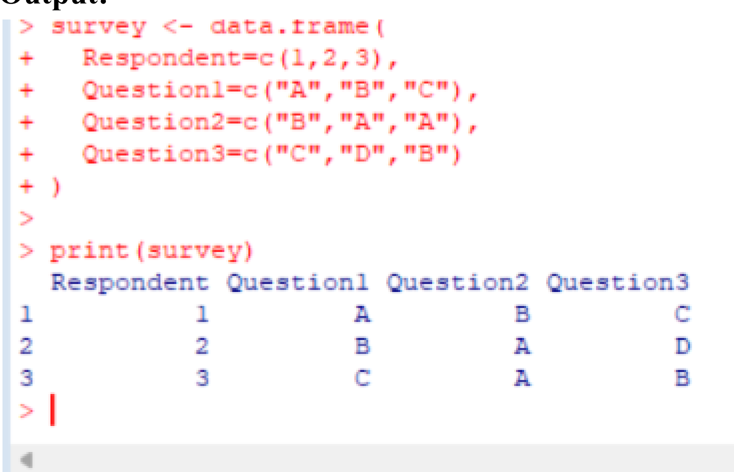


3: Table

Code:

```
survey <- data.frame(
  Respondent=c(1,2,3),
  Question1=c("A","B","C"),
  Question2=c("B","A","A"),
  Question3=c("C","D","B")
)
print(survey)
```

Output:



4: Dashboard (R Shiny)

Code:

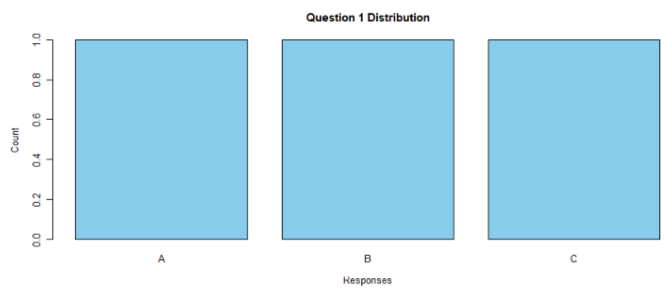
```
library(shiny)
library(fmsb)
```

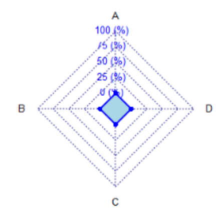
```
survey <- data.frame(
  Respondent=c(1,2,3),
  Question1=c("A","B","C"),
  Question2=c("B","A","A"),
  Question3=c("C","D","B")
)
ui <- fluidPage(
  titlePanel("Survey Results Dashboard"),
  fluidRow(
    column(6, plotOutput("bar")),
    column(6, plotOutput("radar"))
  ),
  br(),
  tableOutput("table")
)
server <- function(input, output){
  output$bar <- renderPlot({
    q1 <- table(survey$Question1)
    barplot(rbind(q1),
            col="skyblue",
            main="Question 1 Distribution",
            xlab="Responses",
            ylab="Count")

  })
  output$radar <- renderPlot({
    radar <- data.frame(
      A=c(3,1,0),
      B=c(3,1,0),
      C=c(3,1,0),
      D=c(3,1,0)
    )
    rownames(radar) <- c("Max","Min","Responses")
    radarchart(radar,
               axistype=1,
               pcol="blue",
               pfcol="lightblue",
               plwd=2)

  })
```

```
  output$table <- renderTable({
    survey
  })
}
shinyApp(ui, server)
```

Output:

Survey Results Dashboard





Respondent	Question1	Question2	Question3
1.00	A	B	C
2.00	B	A	D
3.00	C	A	B

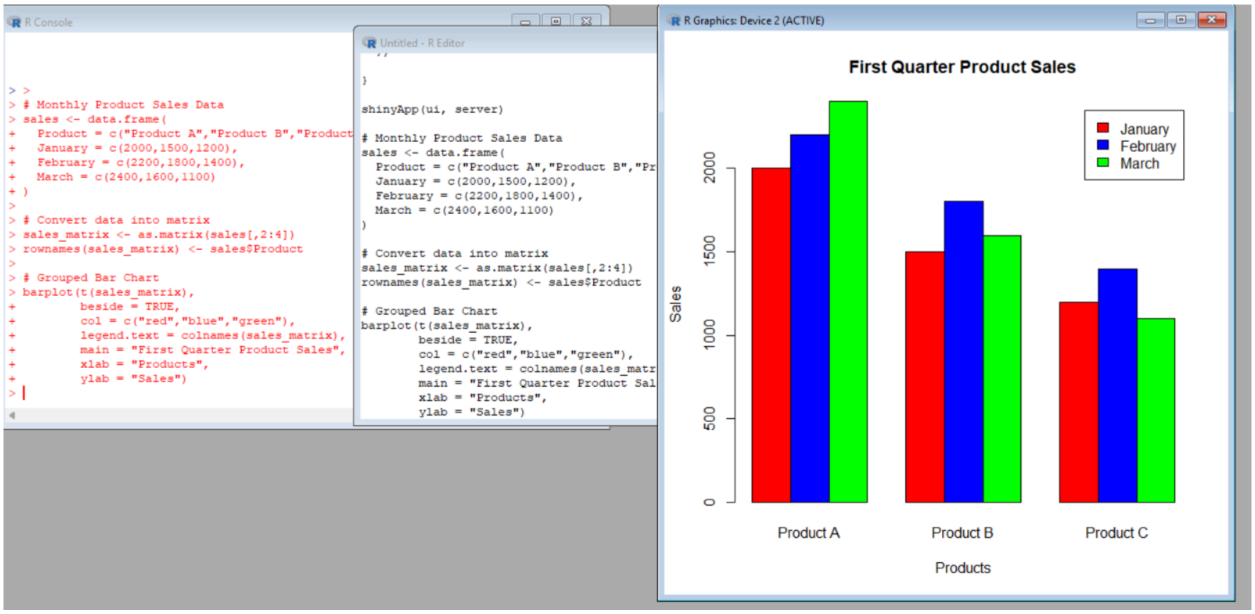
Set 15: Product Sales Analysis

1: Grouped Bar Chart

Code:

```
sales <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)
sales_matrix <- as.matrix(sales[,2:4])
rownames(sales_matrix) <- sales$Product
barplot(t(sales_matrix),
        beside = TRUE,
        col = c("red","blue","green"),
        legend.text = colnames(sales_matrix),
        main = "First Quarter Product Sales",
        xlab = "Products",
        ylab = "Sales")
```

Output:

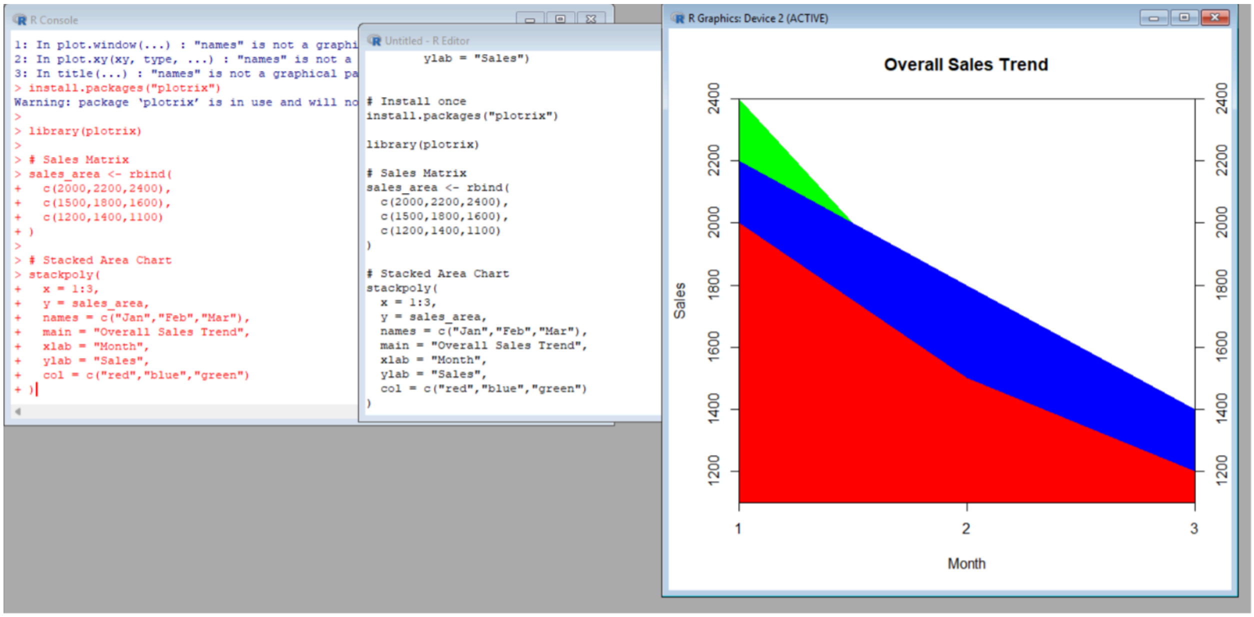


2: Stacked Area Chart

Code:

```
install.packages("plotrix")
library(plotrix)
sales_area <- rbind(
  c(2000,2200,2400),
  c(1500,1800,1600),
  c(1200,1400,1100)
)
stackpoly(
  x = 1:3,
  y = sales_area,
  names = c("Jan","Feb","Mar"),
  main = "Overall Sales Trend",
  xlab = "Month",
  ylab = "Sales",
  col = c("red","blue","green")
)
```

Output:

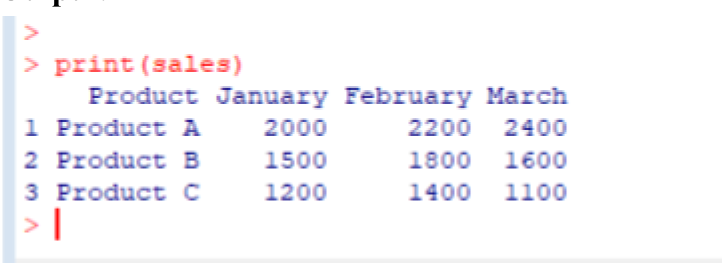


3: Table

Code:

```
sales <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  JanuarySales = c(2000,1500,1200),
  FebruarySales = c(2200,1800,1400),
  MarchSales = c(2400,1600,1100)
)
print(sales)
```

Output:



4: Dashboard (R Shiny)

Code:

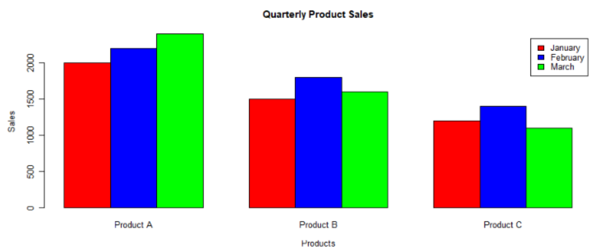
```
library(shiny)
sales <- data.frame(
  Product = c("Product A","Product B","Product C"),
```

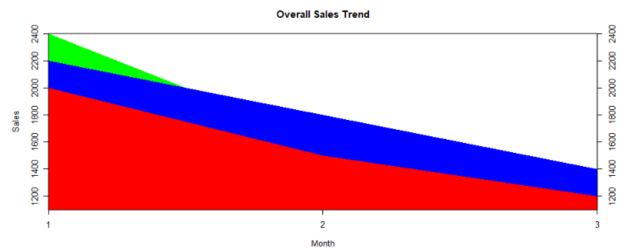
```
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)
ui <- fluidPage(
  titlePanel("Product Sales Dashboard"),
  fluidRow(
    column(6, plotOutput("bar")),
    column(6, plotOutput("area"))
  ),
  br(),
  tableOutput("table")
)
server <- function(input, output){
  output$bar <- renderPlot({
    sales_matrix <- as.matrix(sales[,2:4])
    rownames(sales_matrix) <- sales$Product
    barplot(t(sales_matrix),
            beside = TRUE,
            col = c("red","blue","green"),
            legend.text = colnames(sales_matrix),
            main = "Quarterly Product Sales",
            xlab = "Products",
            ylab = "Sales")
  })
  output$area <- renderPlot({
    library(plotrix)
    sales_area <- rbind(
      c(2000,2200,2400),
      c(1500,1800,1600),
      c(1200,1400,1100)
    )
    stackpoly(
      x = 1:3,
      y = sales_area,
      names = c("Jan","Feb","Mar"),
      col = c("red","blue","green"),
      main = "Overall Sales Trend",
      xlab = "Month",
      ylab = "Sales"
```

```
    )
  })
  output$table <- renderTable({
    sales
  })
}
shinyApp(ui, server)
```

Output:

Product Sales Dashboard





Product	January	February	March
Product A	2000.00	2200.00	2400.00
Product B	1500.00	1800.00	1600.00
Product C	1200.00	1400.00	1100.00